

Professional Cloud Architect Practice Exam Questions

Scenario-Based Multiple-Choice Questions with Grounded Solutions & Case Study Context

Question 1

FOCUS AREA / CASE STUDY: EHR HEALTHCARE CASE STUDY

Scenario: EHR Healthcare is migrating its containerized customer-facing SaaS applications from expiring colocation facilities to Google Cloud. The executive leadership requires a minimum of 99.9% availability for all customer-facing systems, continuous deployment capabilities, reduced latency for global healthcare providers, and minimized infrastructure administration costs. Which architecture best satisfies these business and technical requirements?

- A. Deploy containerized applications to a single zonal Google Kubernetes Engine (GKE) cluster in us-central1, using Cloud Storage for persistent database caching and Standard Network Tier for egress traffic.
- B. Deploy regional GKE clusters across multiple Google Cloud regions behind a global Application Load Balancer using Premium Network Tier, and utilize managed database services with automated regional failover.
- C. Rehost containerized workloads on Compute Engine virtual machines in a single region, managing Kubernetes clusters manually with custom shell scripts and email alerts.
- D. Migrate all application workloads directly to Compute Engine Spot VMs with local SSDs, relying on manual nightly snapshots to Cloud Storage for disaster recovery.

Correct Answer: B

Explanation: Deploying regional GKE clusters across multiple regions behind a global Application Load Balancer guarantees high availability (exceeding the 99.9% SLA target) and routes global users over Google's low-latency Premium Network Tier. Leveraging fully managed database services minimizes operational complexity and administrative overhead while ensuring automated failover during regional disruptions.

Question 2

FOCUS AREA / CASE STUDY: ALTOSTRAT MEDIA CASE STUDY

Scenario: Altostrat Media is modernizing its hybrid media content management platform. They need to ingest high-volume audio and video files securely and with high performance from their legacy on-premises facilities into Google Cloud, while optimizing cloud storage costs for a rapidly expanding media archive. What solution should a Cloud Architect recommend?

- A. Connect the on-premises datacenter using HA VPN over the public internet, and store all media indefinitely in the Standard Cloud Storage class.
- B. Establish Dedicated Cloud Interconnect for secure, high-performance hybrid data ingestion, and implement Cloud Storage Lifecycle Management rules to transition older media assets to Coldline or Archive storage classes.
- C. Ingest media assets using public HTTP REST endpoints into Filestore instances, and manually delete legacy assets every 30 days.
- D. Set up VPC Network Peering between the on-premises network and a Google Cloud VPC, storing all files directly in BigQuery tables as JSON blobs.

Correct Answer: B

Explanation: Dedicated Cloud Interconnect provides the dedicated, high-capacity, low-latency physical link required for continuous, high-throughput media ingestion from on-premises facilities. Object Lifecycle Management in Cloud Storage automatically moves infrequently accessed media files to lower-cost storage classes (Coldline/Archive), satisfying Altostrat's core business requirement to optimize storage spending without sacrificing scalability.

Question 3

FOCUS AREA / CASE STUDY: CYMBAL RETAIL CASE STUDY

Scenario: Cymbal Retail is implementing a generative AI commerce initiative. They want web and mobile app customers to discover products through natural language interactions. Additionally, their technical requirements dictate that retail associates must have a user interface to review, modify, or approve AI-generated catalog attributes before they update the live product catalog. Which combination of Google Cloud capabilities addresses these needs?

- A. Integrate Google Cloud's Discovery AI for natural language product discovery, and deploy a custom Human-in-the-Loop (HITL) review UI for retail associates to manage AI-generated catalog updates.
- B. Replace relational databases with Firestore and run SQL transformations directly inside BigQuery ML without human intervention.
- C. Route customer natural language requests through an Interactive Voice Response (IVR) system to call center agents for manual catalog searching.
- D. Export product catalogs to flat CSV files in Cloud Storage FUSE and rely on open-source Nagios alerts to notify associates of catalog discrepancies.

Correct Answer: A

Explanation: Cymbal Retail's solution concept specifically mandates integrating Google Cloud Discovery AI to process natural language user requests and deliver intuitive product recommendations. Furthermore, their technical requirements explicitly call for a Human-in-the-Loop (HITL) associate review UI to allow staff to approve, reject, or edit generated attributes before syncing changes to the catalog.

Question 4

FOCUS AREA / CASE STUDY: KNIGHTMOTIVES AUTOMOTIVE CASE STUDY

Scenario: KnightMotives is expanding its autonomous vehicle testing program into international markets, including the European Union. They must adhere strictly to EU data protection regulations (such as GDPR), prevent telemetry data exfiltration, ensure data sovereignty, and protect against security breaches. How should the Cloud Architect secure the telemetry data environment?

- A. Store telemetry data in public Cloud Storage buckets and grant public read access to international testing partners to lower network configuration complexity.
- B. Configure VPC Service Controls to establish a security perimeter around sensitive data services, enforce Customer-Managed Encryption Keys (CMEK) via Cloud KMS, and restrict data residency to specific EU regions.
- C. Stream telemetry directly over unencrypted HTTP endpoints into a single Compute Engine VM in us-central1 using static default service account keys.
- D. Rely on legacy SFTP batch scripts to transfer vehicle logs back to an on-premises mainframe, bypassing cloud security perimeter controls.

Correct Answer: B

Explanation: VPC Service Controls prevent unauthorized data exfiltration by restricting API access to sensitive Google Cloud resources. Customer-Managed Encryption Keys (CMEK) via Cloud KMS give the customer complete control over cryptographic key rotation and access policies. Restricting data residency to specific geographical boundaries within the EU satisfies strict regulatory compliance and data sovereignty laws.

Question 5

FOCUS AREA / CASE STUDY: DEVOPS, SECURITY & IAC CORE OBJECTIVES

Scenario: An enterprise cloud architect is designing a secure continuous integration and continuous deployment (CI/CD) pipeline for containerized microservices running on Cloud Run and GKE. The security team mandates that only verified, vulnerability-scanned container images generated by approved build pipelines can be deployed to production. Which architecture implements these guardrails?

- A. Use Cloud Build for automated container compilation, Artifact Analysis to scan images for known vulnerabilities, and Binary Authorization to enforce attestation policies at deployment time.
- B. Allow developers to push container images directly from local workstations to production GKE clusters using kubectl with cluster admin permissions.
- C. Host container images in public Cloud Storage buckets and disable IAM authentication on production cluster worker nodes.
- D. Build container images manually on Compute Engine virtual machines and bypass automated vulnerability scanning to accelerate release velocity.

Correct Answer: A

Explanation: Binary Authorization acts as a deploy-time security gateway on GKE and Cloud Run, ensuring that only trusted container images that carry valid signatures/attestations (such as passing vulnerability scans from Artifact Analysis) are allowed to deploy. Combining Cloud Build, Artifact Analysis, and Binary Authorization embeds shift-left security controls and maintains Supply-chain Levels for Software Artifacts (SLSA) compliance.

PCA Exam Strategy Tip:

On the Professional Cloud Architect exam, prioritize answers that align with **Google Cloud Managed Services**, **High Availability across regions**, **Zero-Trust Security (VPC Service Controls, CMEK, Binary Authorization)**, and **Automated Governance** over manual or custom-coded solutions.